LEO Orbit Dynamics Simulator & PINN Surrogate

Honours Research Briefing Document & Thesis Outline | Meeting with Prof. G. Ferrer

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1. Executive Summary & Core Research Objectives

This investigation presents a high-fidelity astrodynamics simulation suite coupled with an edge-deployable **Physics-Informed Neural Network (PINN) / Deep Learning Surrogate** for Low Earth Orbit (LEO) satellites. Ground truth is generated via high-order adaptive numerical integration (Runge-Kutta Dormand-Prince `RK45`) incorporating geopotential zonal harmonics (J_2) and dynamic thermospheric aerodynamic drag. The ML surrogate model is trained on non-dimensionalized transition state pairs to enable real-time orbit propagation on resource-constrained satellite flight computers (e.g. ESP32 microcontrollers).

2. Meeting Talking Points for Prof. Ferrer

- **Mathematical Rigour & Integrator Baseline:** Ground truth trajectories are generated strictly using high-order variable step-size numerical methods (DOP853/RK45) with local error tolerances of 10^{-9}.
- **Dual Perturbation Coupling:** Simultaneously models conservative gravitational J_2 field effects (nodal precession and apsidal rotation) and non-conservative atmospheric drag (co-rotating atmosphere with altitude-dependent scale heights).
- **Surrogate Paradigm Shift (Residual Transition Map):** Rather than learning absolute whole-trajectory positions (which fail to generalize across multi-orbit rollouts), the neural network learns a one-step *residual velocity/position flow field*: $\mathbf{S}_{t+\Delta t} = \mathbf{S}_t + N_\theta(\mathbf{S}_t)$. This shrinks network size to <10k parameters (≈ 36 KB fp32, ≈ 9 KB int8), maintaining <5% rollout drift over 24 hours.
- **Physics-Informed Loss Enforcement:** Employs PyTorch auto-differentiation (`torch.autograd`) to penalize violations of Newton's second law directly in backpropagation.
- **Automated Parametric Sweep Engine:** Supports automated Monte Carlo and parametric sweeps across ballistic coefficients ($B = m / (C_d A)$), orbital inclination, eccentricity, and perigee altitudes.

3. Thorough Theoretical Formulation & Governing Equations

3.1. Fundamental Equations of Motion (Two-Body with Perturbations)

In the Earth-Centered Inertial (ECI J2000) frame, the state vector $\mathbf{S}(t) = [\mathbf{r}(t)^T, \mathbf{v}(t)^T]^T \in \mathbb{R}^6$ evolves according to:

$$d\mathbf{r}/dt = \mathbf{v}$$

$$d\mathbf{v}/dt = -(\mu_\oplus / \|\mathbf{r}\|^3)\,\mathbf{r} + \mathbf{a}_{J2}(\mathbf{r}) + \mathbf{a}_{drag}(\mathbf{r}, \mathbf{v}) + \mathbf{a}_{3B}(\mathbf{r}, t)$$

where $\mu_\oplus = 3.986004418 \times 10^{14}\ m^3/s^2$ is Earth's geocentric standard gravitational parameter.

3.2. Geopotential Oblateness (J_2) Zonal Harmonic

Earth's equatorial oblateness generates a non-spherical gravitational acceleration field derived from the gradient of the zonal gravitational potential U_{J2}:

$$\mathbf{a}_{J2}(\mathbf{r}) = -(3/2)\cdot(J_2\,\mu_\oplus\,R_\oplus^{\,2}/r^4)\cdot\left[\,(x/r)(1-5\,z^2/r^2),\ \ (y/r)(1-5\,z^2/r^2),\ \ (z/r)(3-5\,z^2/r^2)\,\right]^T$$

Analytical Secular Rates for Model Validation: The secular drift rate of the Right Ascension of the Ascending Node ($d\Omega/dt$) and Argument of Perigee ($d\omega/dt$) serve as analytical validation truth:

$$d\Omega/dt = -(3/2)\,\bar{n}\,J_2\,(R_\oplus/p)^2\cos i$$

$$d\omega/dt = +(3/4)\,\bar{n}\,J_2\,(R_\oplus/p)^2\,(5\cos^2 i - 1)$$

where $\bar{n} = \sqrt{(\mu_\oplus / a^3)}$ is mean motion, $p = a(1-e^2)$ is the semi-latus rectum, $R_\oplus = 6378.137$ km, and $J_2 = 1.08263 \times 10^{-3}$.

3.3. Aerodynamic Drag in a Co-Rotating Thermosphere

The drag acceleration opposes the satellite's velocity vector relative to the rotating atmosphere ($\mathbf{v}_{rel}$):

$$\mathbf{v}_{rel} = \mathbf{v} - \boldsymbol{\omega}_\oplus \times \mathbf{r}$$

$$\mathbf{a}_{drag} = -(1/2)\cdot\rho(h)\cdot(C_dA/m)\cdot\|\mathbf{v}_{rel}\|\,\mathbf{v}_{rel} = -(1/2)\cdot(\rho(h)/B)\cdot\|\mathbf{v}_{rel}\|\,\mathbf{v}_{rel}$$

where $\boldsymbol{\omega}_\oplus = [0, 0, 7.292115 \times 10^{-5}]^T$ rad/s, $B = m/(C_dA)$ is the Ballistic Coefficient (kg/m^2), and $\rho(h) = \rho_0 \exp(-(h - h_0)/H)$ with segmented scale heights $H(h)$.

3.4. Non-Dimensionalization & Normalization Scaling

To prevent gradient vanishing and numerical divergence during deep learning backpropagation, canonical units map all variables to O(1):

$$\tilde{r} = \mathbf{r} / R_{scale}, \&quad; \tilde{v} = \mathbf{v} / v_{scale}, \&quad; \tilde{t} = t / T_{scale}$$

(with R_{scale} = 6,378,137 m, v_{scale} = 8,000 m/s, T_{scale} = 5,400 s)

3.5. PINN Loss & Physics Residual Optimization

The composite loss function penalizes empirical data discrepancy alongside differential equation residuals:

$$L_{total} = L_{data} + \lambda_{phys} \cdot L_{physics}$$

$$L_{data} = (1/N) \sum \| \tilde{S}_{k+1} - \hat{S}_{k+1} \|^2$$

$$L_{physics} = (1/N) \sum [\| \partial \hat{r}/\partial t - \hat{v} \|^2 + \| \partial \hat{v}/\partial t - \mathbf{a}_{total}(\hat{r}, \hat{v}) \|^2]$$

4. Quantitative Figures & Visual Proofs for Thesis

Figure ID & Topic	Description & Visual Elements	Quantitative Metric / Thesis Value
Fig 1: RAAN Nodal Regression Validation	Numerical RK45 $\Omega(t)$ trajectory overlaid against analytical first-order J_2 secular regression over 8 hours ($i = 51.6°$).	Evaluates $R^2 > 0.9999$ and confirms osculating sinusoidal sub-orbit oscillations superimposed on the secular trend.
Fig 2: Multi-Force Altitude Decay Comparison	3-curve comparison: Keplerian Two-Body (conservative), J_2-only (periodic altitude fluctuations), and Full Dynamics (J_2 + Drag).	Quantifies perigee decay rate (km/day) and kinetic energy dissipation rate $d\varepsilon/dt = \mathbf{a}_{drag} \cdot \mathbf{v}$.
Fig 3: Six Orbital Elements Evolution	2 × 2 subplot displaying ($a(t)$, $e(t)$, $i(t)$, $\Omega(t)$) under full perturbation regime.	Demonstrates conservation of inclination short-period boundaries and secular semi-major axis erosion.
Fig 4: Ballistic Sweep vs Lifetime	Parametric sweep of $B \in [10, 120]$ kg/m^2 and initial altitude $h \in [200, 500]$ km vs orbital lifetime (days).	Empirical lifetime scaling law: $T_{deorbit} \propto B \cdot \exp(\alpha\, h)$.
Fig 5: PINN vs RK45 Multi-Orbit Drift	24-hour error accumulation plot: $\|\mathbf{r}_{PINN}(t) - \mathbf{r}_{RK45}(t)\|$ vs elapsed time across 16 orbital revolutions.	Validates <5% cumulative positional error threshold and 98.4% compute-time reduction.

5. Proposed Thesis Chapter Breakdown

1. **Chapter 1: Introduction & Research Problem**
   - 1.1 Background on LEO Mega-Constellations & SmallSat Missions
   - 1.2 Computational Bottlenecks of Onboard Orbital Propagation
   - 1.3 Problem Statement: Real-Time Surrogate Modeling via PINNs
   - 1.4 Aims, Objectives, and Scope of Work
2. **Chapter 2: Literature Review & Theoretical Astrodynamics**
   - 2.1 Two-Body Orbital Mechanics & Keplerian State Formulations
   - 2.2 Geopotential Harmonics & Earth Oblateness ($J_2 - J_6$)
   - 2.3 Upper Atmospheric Density Modeling & Aerodynamic Deceleration
   - 2.4 Numerical Integration Techniques (Runge-Kutta Dormand-Prince, Symplectic Methods)
   - 2.5 Physics-Informed Machine Learning in Computational Mechanics
3. **Chapter 3: Numerical Simulator Architecture & Ingestion Pipeline**
   - 3.1 Modular Dynamics Engine (`leo_simulator`)
   - 3.2 Automated Parametric Sweep & Experiment Logging Architecture
   - 3.3 Data Sanitization, Uniform Splitting, and Deterministic Ingestion (`ai/ingest.py`)
   - 3.4 TLE Live Ingestion & SGP4 Oscar Verification

4. **Chapter 4: Physics-Informed Neural Network (PINN) Design & Training**
    - 4.1 Non-Dimensional State-Space Formulation
    - 4.2 Residual Transition Architecture for Microcontrollers
    - 4.3 Loss Function Derivation: Data Loss, Momentum Residual, and Autograd Formulation
    - 4.4 Training Dynamics, Early Stopping, and Learning Rate Schedules
5. **Chapter 5: Results, Verification & Parametric Sensitivity Analysis**
    - 5.1 Benchmark Verification against Analytical Perturbation Theory
    - 5.2 Orbital Lifetime Analysis across Ballistic Coefficients & Altitudes
    - 5.3 Conjunction Detection & Orbital Collision Impulse Resolution
    - 5.4 PINN Rollout Drift & Computational Speedup Analysis
6. **Chapter 6: Hardware-in-the-Loop / TinyML Feasibility (ESP32 Port)**
    - 6.1 Model Quantization (Float32 to Int8) & Memory Footprint (< 200 KB RAM)
    - 6.2 Execution Latency per Inference Step on Edge Microcontrollers
7. **Chapter 7: Conclusions & Future Recommendations**